

Name of the Student	
Reg. No	
GitHub Link	

SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 2

Game Based Learning

COURSE NAME: DEEP LEARNING

COURSE CODE: MLA04

SLOT :

COURSE OUTCOME COVERED

CO3: Analyze and apply convolutional neural network architectures, including convolutional layers, filters, parameter sharing, regularization, AlexNet and ResNet, to develop solutions for image classification and real-world computer vision applications. (L4)

CO Weightage: 25%

Duration: 1 Hour

Assessment Tool Weightage: 35%

Marks: 50

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Name of the student:

Criteria	Understanding of Concept (2.5)	Conceptual Clarity (2.5)	Problem-Solving Ability (2)	Solution Design & Application (2)	Teamwork & Game Participation (1)	Total (10)
Weightage	25 %	25%	20%	20 %	10 %	100%
Round 1						
Round 2						
Total						

Signature of the Faculty:

Total Marks :

QUESTIONS:01

Total Marks: 50

Question:

Design and participate in a two-round game-based learning activity to explore the fundamental concepts of Convolutional Neural Networks (CNNs), including architecture, layers, filters, parameter sharing, regularization, AlexNet, ResNet, and real-world applications. Progress through five levels in each round, applying your knowledge to solve challenges involving CNN design, feature extraction, architecture comparison, and practical applications.

The Game Progression

Round 1:

Basic CNN Concepts → Filters → Parameter Sharing → Architecture → AlexNet vs ResNet

Round 2:

Convolution → Feature Maps → Regularization → Architecture Comparison → Real-World Application

ROUND 1: CNN Architecture Builder (Learn by constructing and understanding CNN architectures layer by layer)

Level	Challenge	Description
Level 1	CNN Layer Matcher	Match CNN layers such as Convolution, Pooling, Flatten, and Fully Connected layers with their respective functions and roles in the network.
Level 2	Filter Finder	Identify suitable filters and kernels for feature extraction and understand how filter size, stride, and padding affect the output feature map.
Level 3	Parameter Sharing Puzzle	Determine how the same filter weights are reused across different image locations and calculate the number of parameters in a convolutional layer.
Level 4	CNN Architecture Debugger	Identify and correct problems in a CNN architecture, such as incorrect layer order, incompatible dimensions, and inappropriate layer selection.
Level 5	AlexNet vs ResNet Challenge	Compare AlexNet and ResNet architectures and identify how their architectural approaches differ, particularly the use of residual connections in ResNet.

ROUND 2: CNN Feature Detective (Learn by exploring how CNNs extract features, avoid overfitting, and solve real-world problems)

Level	Challenge	Description
Level 1	Edge Detection Explorer	Apply simple (3x3) filters to an image and determine how convolution operations can detect edges and other basic features.
Level 2	Feature Map Detective	Analyze feature maps produced by different CNN layers and identify the progression from simple features such as edges to complex patterns and objects.
Level 3	Regularization Rescuer	Identify overfitting in a CNN model and select suitable regularization techniques to improve generalization and model performance.

Level	Challenge	Description
Level 4	CNN Architecture Race	Compare ResNet and AlexNet for different image-classification applications and determine which architecture is more suitable for a given problem.
Level 5	CNN Application Master	Design a CNN-based solution for a real-world application such as medical image analysis, face recognition, object detection, or autonomous driving, and justify the selected architecture and layers.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Conceptual Clarity	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Problem-Solving Ability	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design & Application	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Teamwork & Game Participation	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand